

ORIGINAL ARTICLE

Orthostatic Blood Pressure, Cardiovascular Disease, and Hypotensive Events

Manfred N. Mate-Kole¹, Mingyu Zhang², Ruth-Alma N. Turkson-Ocran³, Fredrick L. Kwabong⁴, B. Gwen Windham, Elizabeth Selvin⁵, Pamela L. Lutsey⁶, Stephen P. Juraschek⁷

BACKGROUND: Orthostatic hypotension is thought to be associated with coronary heart disease, falls, and syncope due to low blood pressure (BP) upon standing.

METHODS: The ARIC (Atherosclerosis Risk in Communities) study measured supine and standing BP among adult participants aged 45 to 64 years once at baseline and followed them for over 35 years. We evaluated higher and lower supine and standing systolic BP, diastolic BP, mean arterial pressure, pulse pressure, absolute and relative orthostatic changes in BP after standing, and mean BP across positions. Associations with adjudicated coronary heart disease and mortality events, as well as hospitalizations and medical claims-based falls and syncope, were assessed via adjusted Cox models in strata of antihypertensive treatment.

RESULTS: Among 11 386 participants (mean age, 54 years [SD, 5.7 years]; 56% female; 25% Black adults), drops in systolic BP upon standing (absolute or relative) were associated with coronary heart disease, syncope, and mortality. Higher supine systolic BP and mean arterial pressure were associated with syncope among untreated participants. Increases in systolic BP ≥ 20 mmHg upon standing were associated with falls (hazard ratio, 1.52 [95% CI, 1.14–2.02]) and syncope (hazard ratio, 1.40 [95% CI, 1.03–1.92]), particularly among untreated participants. Lower standing systolic BP was associated with a higher risk of syncope among treated participants (hazard ratio, 1.55 [95% CI, 1.14–2.12]). Regardless of treatment status, a higher pulse pressure was associated with coronary heart disease and mortality, but this was not observed for falls or syncope.

CONCLUSIONS: Higher BP, rather than lower standing BP alone, may be an important risk factor for both cardiovascular and hypotension-related events, especially among untreated adults. (*Hypertension*. 2026;83:2153–2167. DOI: 10.1161/HYPERTENSIONAHA.125.25773.) • **Supplement Material.**

Key Words: accidental falls ■ blood pressure ■ cardiovascular diseases ■ death ■ humans ■ hypotension, orthostatic ■ syncope

Extreme blood pressure (BP) fluctuations or resting values across different body positions have been associated with adverse cardiovascular outcomes, including cardiovascular disease (CVD), syncope, falls, and death.^{1–4} While most traditional orthostatic BP definitions highlight large absolute drops in BP from supine to standing positions, these conventional definitions may overlook other clinically relevant BP patterns. For example, relative changes may better account for baseline BP, identifying risk from smaller magnitude changes in BP when resting BP is low. Similarly, absolute extremes in

BP across body positions (eg, supine or standing hypertension or hypotension) may be equally or more important for adverse events than changes in BP alone.^{5–7}

Mean arterial pressure (MAP) and pulse pressure are also used to summarize data from both systolic BP (SBP) and diastolic BP (DBP). It is largely viewed that a low MAP is associated with hypotension-related events (eg, falls and syncope), while a high MAP is associated with CVD.^{8–10} In contrast, a large pulse pressure (ie, the difference between SBP and DBP) has been considered a high-risk state for hypotensive events and hypoperfusive

Correspondence to: Stephen P. Juraschek, MD, PhD, Division of General Medicine, Department of Medicine, Beth Israel Deaconess Medical Center, Harvard Medical School, 330 Brookline Ave, CO-1309, #217, Boston, MA 02215. Email sjurasch@bidmc.harvard.edu
Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/HYPERTENSIONAHA.125.25773>.

© 2026 American Heart Association, Inc.

Hypertension is available at www.ahajournals.org/journal/hyp

NOVELTY AND RELEVANCE

What Is New?

Both blood pressure (BP) drops and increases upon standing were associated with hypotension-related events. Unlike lower standing systolic BP, higher pulse pressure was not associated with syncope among hypertensive-treated adults.

What Is Relevant?

High BP may coexist with orthostatic dysregulation and contribute to hypotension-related events. Seated BP alone may miss clinically important patterns in BP across body positions.

Clinical/Pathophysiological Implications?

Orthostatic BP changes may reflect underlying impaired cerebral autoregulation or vascular stiffness that portend vulnerability to cardiovascular and hypotension-related events. Prospective studies are needed to understand causality and define actionable thresholds.

Nonstandard Abbreviations and Acronyms

ARIC	Atherosclerosis Risk in Communities
BP	blood pressure
CHD	coronary heart disease
CVD	cardiovascular disease
DBP	diastolic blood pressure
HDL	high-density lipoprotein
HR	hazard ratio
MAP	mean arterial pressure
SBP	systolic blood pressure

absolute and relative drops in SBP or DBP upon standing would be independently associated with a higher risk of hypotension-related adverse events such as falls and syncope.

METHODS

Data Availability

The data from this study are available by request from the ARIC Coordinating Center and are publicly available through the Biologic Specimen and Data Repository Information Coordinating Center.¹⁹

Study Population

The ARIC study is a community-based, prospective cohort study conducted to evaluate risk factors for subclinical and clinical CVD. The ARIC study enrolled 15 792 adults aged 45 to 64 years between 1987 and 1989 at 4 sites: Forsyth County, North Carolina; Jackson, Mississippi; 8 northern suburbs of Minneapolis, Minnesota; and Washington County, Maryland. The original ARIC study protocol involved physical examinations, medical interviews, laboratory investigations, and, as part of an ancillary study, orthostatic BP measurements. Participants have been followed for over 35 years through periodic clinical visits, active surveillance of the community hospitals, and telephone interviews.^{20,21}

For the present analytic sample, we included participants who attended baseline (visit 1) and excluded participants who withdrew consent (N=35), participants who did not undergo both supine and standing BP measurements at baseline (N=2548), participants with a history of CHD, heart failure, or stroke at baseline (N=1347), and those missing relevant covariate data at baseline (N=476; Figure S1). Our final analytic sample included 11 386 participants. Written informed consent was obtained from all participants, and the ARIC protocol was approved by institutional review boards at all participating study sites. For this secondary analysis of de-identified data, the institutional review board at the Beth Israel Deaconess Medical Center determined the project to be exempt human subjects research.

end-organ injury,^{11–13} such that some experts recommend avoiding hypertension treatment among older adults with a large pulse pressure.^{14,15} Evaluation of BP metrics by treatment status is especially important, as naturally occurring low BP may be protective against adverse events, but low BP in the setting of antihypertensive treatment could represent overtreatment, placing patients at risk for hypotension-related adverse events (eg, falls or syncope).^{16–18} However, there are limited prospective data to inform these traditional perspectives, especially according to antihypertensive treatment status. Using data from the ARIC study (Atherosclerosis Risk in Communities), a large cohort of middle-aged adults, we sought to identify novel and traditional BP metrics associated with CVD or hypotension-related events (falls or syncope), based on standardized supine and standing BP measures by hypertension treatment status. We hypothesized that (1) higher supine and standing SBP, DBP, and MAP would be associated with an increased risk of CVD and all-cause mortality; (2) lower SBP, DBP, and MAP, and higher pulse pressure values would be associated with an increased risk of falls and syncope, especially among treated individuals; and (3) Higher

Exposures: Supine and Standing BP

During the baseline visit, supine and standing BP were measured using a standardized protocol with an automatic oscillometric device (Dinamap 1846 SX) after participants had rested for 20 minutes. The device was programmed to record BP up to 5× at 20- to 30-second intervals over 2 minutes in both positions. At least 4 supine BP measurements were obtained for 90% of participants. The average of all available readings in each position was used to calculate supine and standing BP. Participants were instructed to stand with their arms supported on a bedside table at heart level. Any participant who reported dizziness while standing was allowed to pause briefly in a seated position or lean against the examination table until they felt safe to stand. Otherwise, standing BP was measured immediately after the participant stood, using the same protocol as for supine BP. At least 4 valid standing BP measurements were recorded for 91% of participants. Further details for recording supine and standing BP have been reported previously.²² Note that all BP measurements were obtained during a single visit at baseline and not during the follow-up visits.

We evaluated a comprehensive set of BP patterns including (1) high and low supine BP, (2) high and low standing BP, (3) absolute and relative orthostatic BP changes (orthostatic hypotension and orthostatic hypertension), (4) mean BP across both positions, and (5) derived metrics such as MAP and pulse pressure. High supine and standing BP were defined as SBP ≥ 165 mmHg or DBP ≥ 95 mmHg based on the mean. Low supine and standing BP were defined as a standing SBP ≤ 100 mmHg or DBP ≤ 55 mmHg based on the mean. These thresholds were chosen to represent a similar proportion of the population as orthostatic hypotension (after removing the excluded sample), rounded to the nearest multiple of 5 for ease of clinical translation.

Orthostatic hypotension and orthostatic hypertension were defined by a decrease or increase in SBP ≥ 20 mmHg or DBP ≥ 10 mmHg upon standing, consistent with consensus clinical definitions used for orthostatic hypotension.²³ We categorized supine and standing SBP and DBP changes into relative and absolute thresholds, such as an absolute increase or drop more extreme than ± 20 mmHg or a relative drop or increase more extreme than $\pm 15\%$. Percent cut points were selected to align with population percentiles that approximate a 20 mmHg and 10 mmHg difference between supine and standing SBP and DBP, consistent with traditional orthostatic hypotension thresholds.^{20,21}

We also assessed MAP and pulse pressure in both supine and standing positions based on their relevance as derived metrics of perfusion and vascular function. MAP is traditionally calculated as $(2 \times \text{DBP} + \text{SBP})/3$ and pulse pressure as $(\text{SBP} - \text{DBP})$. We defined high MAP as ≥ 110 mmHg and low MAP as ≤ 70 mmHg, and high pulse pressure as ≥ 80 mmHg and low pulse pressure as ≤ 35 mmHg, also aligning with population percentiles similar to orthostatic hypotension to facilitate comparison.^{24–26}

Outcomes: CHD, Falls, Syncope, and All-Cause Mortality

The primary clinical outcomes were incident CHD, falls, syncope, and all-cause mortality assessed from the baseline visit, with participant follow-up administratively censored

on December 31, 2019, and with follow-up censored on December 31, 2017, for participants at the Jackson site. Hospitalizations among ARIC participants were identified through queries on regular phone calls (annual before 2012 and twice yearly thereafter), surveillance of hospitals in local communities, state and national death indices (with next-of-kin interviews), and linkage to Centers for Medicare and Medicaid Services claims data for Medicare fee-for-service beneficiaries from 1985 to 2018. Falls and syncope were defined as the first occurrence of any relevant hospitalization or health care claim for outpatient or inpatient services following the baseline visit based on *International Classification of Diseases, Ninth Revision* or *International Classification of Diseases, Tenth Revision* codes (Supplemental Methods 1).^{27,28}

Incident CHD events were adjudicated by an expert panel using established protocols. CHD events were determined by a composite definition of fatal CHD, silent myocardial infarction identified by ECG changes, and cardiac procedures. All events underwent adjudication by physicians and trained investigators who evaluated clinical records and death certificates.²⁰

Covariates of Interest

Baseline covariates were collected during visit 1 by trained investigators using standardized protocols with quality control measures. Covariates were selected based on existing literature showing their association with orthostatic BP and cardiovascular and hypotension-related outcomes.^{29,30} Baseline hypertension was based on a seated SBP ≥ 130 mmHg or a seated DBP ≥ 80 mmHg. More severe hypertension was defined by SBP ≥ 140 mmHg or DBP ≥ 90 mmHg (regardless of treatment status). Participants also underwent review of pill bottles, which was also used to determine which participants were taking antihypertensive medications over the past 2 weeks. Participants reported their age, sex, race, education attainment, smoking status, alcohol use, antidepressant medication use, sedative medication use, hypnotic medication use, antipsychotic medication use, anticholinergic medication, and cholesterol-lowering medication use during the prior 4 weeks based on medications brought to the clinic. Race and clinic center were jointly classified as a race-center covariate to reflect the overlap between the cohort's composition and center geographic distribution: White participants from Washington County, MD; Minneapolis, MN; and Forsyth County, NC; and Black participants from Jackson, MS, and Forsyth County, NC.

Cardiometabolic covariates included body mass index calculated from height and weight and estimated glomerular filtration rate calculated using the 2021 Chronic Kidney Disease Epidemiology Collaboration race-free creatinine equation.³¹ Serum HDL (high-density lipoprotein) and total cholesterol were measured using standard laboratory assays. Diabetes was defined by a fasting blood glucose ≥ 126 mg/dL, a non-fasting blood glucose ≥ 200 mg/dL, a self-reported physician diagnosis of diabetes, or diabetes medication use. Heart rate was determined from a baseline ECG taken at rest. Seated BP was based on the average of the second and third of 3 measurements performed using a Hawksley random-zero sphygmomanometer. Measurements were performed by a trained technician after 5 minutes of seated rest and separated by 1-minute intervals. Leisure-time physical activity was assessed using the validated ARIC/Baecke Physical Activity Questionnaire.³²

Statistical Analyses

Baseline study population characteristics were described using means and proportions overall and according to hypertension treatment status for both the included and excluded populations.

We evaluated the association of supine and standing hypotension, standing and supine hypertension, mean SBP and DBP across both positions, and absolute and relative changes in SBP and DBP after standing with CHD, all-cause mortality, falls, and syncope. We used Cox proportional hazards models to estimate hazard ratios (HRs) and 95% CIs in the overall population and in strata of hypertension treatment status.

Models were adjusted for age, race-center, estimated glomerular filtration rate, body mass index, heart rate, HDL cholesterol, total cholesterol, diabetes, cholesterol-lowering medication use (yes/no), alcohol use (never, former, or current), education attainment (<high school completion, high school degree or equivalent or vocational school, or at least some college), leisure activity index, and smoking status (never, former, or current).

We repeated models stratified by hypertension treatment status and included interaction terms between exposures and hypertension treatment status to assess potential effect modification. We also evaluated associations between a number of other supine and standing BP metrics, including high and low MAP, high and low pulse pressure, categories of hypertension (SBP ≥ 140 mmHg or DBP ≥ 90 mmHg), and categories of hypotension (SBP ≤ 105 mmHg or DBP ≤ 60 mmHg) with outcomes. Models were also evaluated in strata of sex (male or female), self-reported dizziness (yes/no), antihypertensive medication class (angiotensin-converting enzyme inhibitor, β -blocker, calcium channel blocker, loop diuretic, or thiazide diuretic), and more severe hypertension (SBP ≥ 140 mmHg/DBP ≥ 90 mmHg versus SBP < 140 mmHg and DBP < 90 mmHg).

In secondary analyses, distinct Cox regression models were used to examine the independent associations of categories of change in positional SBP and DBP (per 10 mmHg and 5 mmHg increments/decrements, respectively), as well as categories of postural SBP change (per 10 mmHg increments and decrements) and postural DBP change (per 5 mmHg increments and decrements), with outcomes. Given the number of BP metrics and outcomes evaluated, we acknowledge the potential for false-positive findings due to multiple comparisons. However, we chose not to apply formal correction methods (eg, Bonferroni or false discovery rate procedures) because our primary goal was to explore patterns of association and assess the robustness of findings across related exposures and outcomes.

We used kernel density plots to visualize the distribution of BP values and the continuous relationship between BP changes and outcomes. We also used restricted cubic splines with knots positioned using the Harrell method to visualize potential nonlinear associations between postural BP parameters and clinical outcomes.³³ Splines were implemented overall and in strata of treatment status. The linearity assumption was evaluated using Wald tests comparing nested models of the continuous linear metric with cubic spline terms to models without the spline terms.

All statistical analyses were conducted using Stata 15.1 (StataCorp, College Station, TX). A 2-tailed $P < 0.05$ was considered statistically significant.

RESULTS

Population Characteristics

Among 11386 participants, the mean age of participants was 54 years (SD, 5.7 years; range, 44–66 years); 56% were female, 25% were Black adults, and 25% were treated for hypertension. During the baseline BP assessment, 10% of participants self-reported dizziness upon standing. Supine and standing SBP and DBP were higher among participants treated with antihypertensive medications compared with those who were untreated. Elevated seated BP ($\geq 130/\geq 80$ mmHg) was found among 55% of hypertension-treated participants and 31% of untreated participants (Table 1; [Tables S1 through S3](#)). The distribution of BP values by position, orthostatic changes, MAP, and pulse pressure is shown in [Figure S2](#).

Participants were followed for a median of 25 to 28 years (varying by outcome). There were 2448 CHD events, 3230 fall events, 2860 syncope events, and 5840 deaths.

Orthostatic BP and Cardiovascular Outcomes

Among untreated participants, both absolute (≥ 20 mmHg) and relative ($\geq 15\%$) drops in standing SBP were strongly associated with elevated risk of CHD and all-cause mortality (Table 2). An absolute drop of ≥ 20 mmHg in SBP was associated with nearly a 2-fold higher risk of CHD (HR, 1.95 [95% CI, 1.50–2.53]) and death (HR, 1.79 [95% CI, 1.53–2.11]). Similar patterns were observed for DBP drops ([Tables S4 and S5](#)), where a statistically significant association was observed between a ≥ 10 mmHg drop and CHD (HR, 2.17 [95% CI, 1.53–3.08]) or death (HR, 1.73 [95% CI, 1.35–2.23]). There was also a statistically significant association between high supine SBP (≥ 165 mmHg) and higher CHD and mortality risk (HRs > 1.60), with similar findings for high MAP (≥ 110 mmHg) and pulse pressure (≥ 80 mmHg) in supine and standing positions (Table 3; [Table S6](#)). In sex-stratified analyses, female adults with standing SBP drops ≥ 20 mmHg had a higher risk of CHD and mortality than male adults ([Tables S7 through S10](#)). Continuous spline models were generally linear except for all-cause mortality (Figure; [Figures S3 and S4](#); [Table S11](#)). Similar patterns were observed for DBP ([Figure S5](#)). Associations of supine and standing MAP were similar to those of SBP and DBP, with supine MAP strongly associated with risk across all outcomes ([Figure S6](#)).

Among treated participants, the overall direction of observed associations was similar, although slightly attenuated (Table 4). There remained a statistically significant association between larger drops in SBP (≥ 20 mmHg) and CHD (HR, 1.82 [95% CI, 1.44–2.30]) and mortality (HR, 1.46 [95% CI, 1.24–1.73]). Treated individuals with supine SBP ≥ 165 mmHg or MAP ≥ 110 mmHg also

Table 1. Baseline Characteristics Overall and by Hypertension Treatment Status, Mean (SD) or %

Characteristic	Overall, N=11 386	Untreated, N=8524	Treated, N=2862
Age, y	53.9 (5.7)	53.5 (5.7)	55.3 (5.7)
Female, %	56	54	62
Black, %	25	21	38
Race-study center, %			
Washington County (White)	25	25	26
Jackson (Black)	22	18	34
Minneapolis (White)	26	28	20
Forsyth County (Black)	3	3	4
Forsyth County (White)	24	26	16
Seated SBP, mm Hg	120.8 (18.6)	118.2 (17.5)	128.4 (19.7)
Seated DBP, mm Hg	73.4 (11.0)	72.2 (10.6)	77.1 (11.5)
Supine SBP, mm Hg	124.6 (19.6)	121.9 (18.1)	132.7 (21.5)
Supine DBP, mm Hg	72.3 (9.7)	71.5 (9.3)	74.6 (10.4)
Standing SBP, mm Hg	124.7 (20.0)	122.4 (18.7)	131.5 (21.9)
Standing DBP, mm Hg	75.5 (10.4)	74.9 (10.0)	77.5 (11.2)
Seated BP $\geq 130/\geq 80$ mm Hg	37	31	55
More severe HTN (SBP ≥ 140 mm Hg or DBP ≥ 90 mm Hg)	17	13	29
Heart rate, bpm	66.6 (10.0)	66.6 (9.8)	66.5 (10.8)
Body mass index, kg/m ²	27.4 (5.2)	26.8 (4.8)	29.2 (5.8)
Estimated glomerular filtration rate, mL/min per 1.73 m ²	102.0 (12.7)	103.3 (11.2)	98.2 (15.7)
HDL cholesterol, mg/dL	52.2 (17.3)	52.8 (17.5)	50.4 (16.5)
Total cholesterol, mg/dL	214.0 (41.3)	212.2 (40.0)	219.1 (44.4)
Cholesterol-lowering medication use, %	3	2	5
Antihypertensive use, %			
ACE inhibitor	3	0	11
β -Blocker	9	0	35
Calcium channel blocker	2	0	8
Loop diuretic	1	0	5
Thiazide diuretic	7	0	27
Hypnotic medication use, %	2	2	2
Antipsychotic medication use, %	1	0	1
Antidepressant medication use, %	3	2	4
Anticholinergic medication use, %	2	2	2
Sedative medication use, %	2	1	2
Diabetes, %	10	7	19
Physical activity level*	2.4 (0.6)	2.4 (0.6)	2.3 (0.6)
Dizziness upon standing, %	10	9	11
Alcohol use, %			
Never	25	23	31
Former	18	17	20
Current	58	61	49
Education level, %			
Less than completed high school	22	19	29
High school degree or equivalent or vocational school	42	42	40
At least some college	37	39	31
Smoking status, %			
Never	43	41	46

(Continued)

Table 1. Continued

Characteristic	Overall, N=11 386	Untreated, N=8524	Treated, N=2862
Former	32	32	32
Current	26	27	23

*Range of 1–5 with 1 representing least active and 5 representing most active. ACE indicates angiotensin-converting enzyme; DBP, diastolic blood pressure; HDL, high-density lipoprotein cholesterol; HTN, hypertension; and SBP, systolic blood pressure.

had a statistically significant mortality risk (HRs >1.65; Tables 4 and 5). In contrast, low mean supine and standing BP (≤ 100 mmHg), which was inversely associated with CHD, syncope, and mortality in untreated adults, was not associated with any events among the treated group (Tables S3 and S4). We repeated these analyses in strata of orthostatic dizziness with similar results (Tables S12 through S15).

Hypotension-Related Outcomes: Falls and Syncope

Orthostatic SBP changes in either direction (drops and increases) were associated with a higher risk of falls and syncope among untreated individuals. A ≥ 20 mmHg drop in SBP after standing was associated with a higher risk of syncope (HR, 1.40 [95% CI, 1.06–1.85]) and falls (HR, 1.36 [95% CI, 1.04–1.77]), while a ≥ 20 mmHg rise in SBP was also associated with a higher risk of falls (HR, 1.52 [95% CI, 1.14–2.02]) and syncope (HR, 1.40 [95% CI, 1.03–1.92]; Table 2). Composite definitions of orthostatic hypertension, particularly an increase in SBP ≥ 20 mmHg or DBP ≥ 10 –15 mmHg, were associated with falls (HR, 1.37 [95% CI, 1.09–1.72]) and syncope (HR, 1.31 [95% CI, 1.02–1.68]; Table 3). Standing SBP drops ≥ 20 mmHg were associated with higher syncope risk among both sexes, with slightly stronger associations in female adults than male adults. Supine SBP ≥ 165 mmHg was also associated with higher syncope risk among female adults (Tables S7 through S10).

We observed similar patterns among treated participants. A ≥ 20 mmHg drop in SBP was associated with a higher syncope risk (HR, 1.48 [95% CI, 1.14–1.92]), and a drop in DBP ≥ 10 mmHg was associated with a higher risk of falls (HR, 1.59 [95% CI, 1.03–2.45]) and syncope (HR, 1.87 [95% CI, 1.21–2.88]; Table S5). Continuous spline models also demonstrated that absolute DBP drops were generally linear except for syncope and all-cause mortality (Figure S7; Table S11). Relative (percent) DBP drops also showed nonlinear associations with syncope (Figure S8). A low standing SBP (≤ 100 mmHg) was associated with higher syncope risk among treated adults (HR, 1.55 [95% CI, 1.14–2.12]), while no such association was seen among the untreated group (Tables 2 and 4). Pulse pressure, including a pulse pressure ≥ 80 mmHg, was not associated with falls or syncope in either group (Tables 3 and 5; Table S6).

Continuous spline models showed generally linear associations between pulse pressure and outcomes except for all-cause mortality (Figure S9).

Treatment Interactions and Categorical Changes in BP

Across treatment groups, the magnitude and direction of observed associations between orthostatic BP changes and clinical outcomes varied minimally by antihypertensive treatment status (Tables S16 and S17). Compared with untreated participants, treated individuals demonstrated weaker associations between standing SBP ≤ 100 mmHg and CHD (P interaction <0.05), and between changes in SBP upon standing ≥ 20 mmHg and death (P interaction <0.05). However, treatment, antihypertensive medication class, and hypertension severity did not modify the association between orthostatic drops and CHD (Tables 2 and 3; Tables S18 through S41).

Categorical analyses using absolute BP levels and changes confirmed prior observations (Table S42). Compared with reference categories, higher supine and standing SBP (≥ 160 mmHg) and DBP (≥ 90 mmHg) were consistently associated with higher risk of CHD and all-cause mortality, regardless of treatment status (Tables S43 and S44). Postural drops in SBP (≥ 20 mmHg) or DBP (≥ 10 mmHg) were associated with higher risk of CHD, syncope, and mortality.

DISCUSSION

Among this large cohort of middle-aged Black and White adults, we found that elevated SBP and orthostatic decreases in SBP were associated with a higher risk of CHD and all-cause mortality. Moreover, both increases and decreases in SBP upon standing (absolute or relative) were associated with a higher risk of falls and syncope among untreated participants. Risk associations were largely consistent across treated and untreated participants, with the exception of low standing BP, which was associated with a higher risk of syncope among the treated population. These findings challenge traditional paradigms that view hypertension and hypotension as separate entities, suggesting instead that hypertension itself may contribute to hypotension-related events.

Elevated BP values, whether supine, standing, or averaged across positions, were strongly and consistently associated with increased risk of CHD and all-cause

Table 2. Association of Different Orthostatic Systolic Blood Pressure Metrics With Cardiovascular and Hypotension-Related Outcomes Among Untreated Adults, N=8524 (HR, 95% CI)

	Proportion of the population, %	HR (95% CI), Coronary heart disease	HR (95% CI), Falls	HR (95% CI), Syncope	HR (95% CI), Mortality
Self-reported dizziness upon standing	9.1	1.11 (0.94–1.31)	1.20 (1.05–1.37)*	1.19 (1.02–1.37)*	1.08 (0.97–1.20)*
Change (standing minus supine)					
Standing SBP drops ≥20 mm Hg†	2.3	1.95 (1.50–2.53)*	1.36 (1.04–1.77)*	1.40 (1.06–1.85)*	1.79 (1.53–2.11)*‡
Standing SBP drops ≥15%	2.3	1.83 (1.41–2.38)*	1.24 (0.95–1.62)	1.45 (1.11–1.90)*	1.60 (1.36–1.89)*
Standing SBP increase ≥20 mm Hg†	1.5	0.82 (0.53–1.26)	1.52 (1.14–2.02)*	1.40 (1.03–1.92)*	1.12 (0.88–1.41)
Standing SBP increase ≥15%	2.6	0.83 (0.59–1.16)	1.17 (0.93–1.47)	1.18 (0.92–1.52)	0.85 (0.70–1.04)
High blood pressure					
Supine SBP ≥165 mm Hg	2.4	1.90 (1.45–2.50)*	0.93 (0.69–1.25)	1.32 (1.01–1.74)*	1.67 (1.42–1.96)*
Standing SBP ≥165 mm Hg	2.7	1.44 (1.10–1.88)*	0.92 (0.69–1.22)	1.09 (0.82–1.44)	1.61 (1.38–1.89)*
Mean supine and standing SBP ≥165 mm Hg	2.2	1.71 (1.27–2.31)*	1.07 (0.78–1.47)	1.12 (0.82–1.53)	1.71 (1.44–2.04)*
Low blood pressure					
Supine SBP ≤100 mm Hg	6.7	0.60 (0.45–0.80)*	0.96 (0.81–1.13)§	0.83 (0.69–1.01)	0.84 (0.72–0.97)*
Standing SBP ≤100 mm Hg	8.0	0.70 (0.56–0.88)*¶	0.95 (0.82–1.10)	0.86 (0.72–1.03)#	0.90 (0.79–1.02)
Mean supine and standing SBP ≤100 mm Hg	6.2	0.57 (0.43–0.76)*	1.01 (0.86–1.19)**	0.78 (0.64–0.96)*††	0.85 (0.73–0.99)*
Continuous (orthostatic change in standing)					
Change in SBP per 10 mm Hg	...	0.89 (0.85–0.94)*	0.95 (0.91–1.00)	0.94 (0.90–0.99)*	0.91 (0.88–0.94)*
Percent change in SBP per 10 percentage points	...	0.88 (0.82–0.94)*	0.95 (0.90–1.00)	0.93 (0.88–0.99)*	0.90 (0.86–0.93)*

Cox proportional hazards models with adjustment for age, race-study center, estimated glomerular filtration rate, body mass index, heart rate, high-density lipoprotein cholesterol, total cholesterol, diabetes, cholesterol-lowering medication use, alcohol use, education attainment, leisure activity index, and smoking status. HR indicates hazard ratio; and SBP, systolic blood pressure.

*Values were statistically significant with a $P<0.05$.
†The 20 mm Hg value is used clinically to define postural change in blood pressure for both orthostatic hypotension and orthostatic hypertension. The other thresholds were chosen to be a similar proportion in the population as orthostatic hypotension ($\approx 2.3\%$) rounded to the nearest multiple of 5 for ease of clinical translation.
‡The P value of the interaction between treatment status and standing SBP drops ≥ 20 mm Hg for mortality was $P=0.035$.
§The P value of the interaction between treatment status and supine SBP ≤ 100 mm Hg for falls was $P=0.013$.
¶The P value of the interaction between treatment status and standing SBP ≤ 100 mm Hg for coronary heart disease was $P=0.004$.
#The P value of the interaction between treatment status and standing SBP ≤ 100 mm Hg for syncope was $P<0.001$.
**The P value of the interaction between treatment status and mean supine and standing SBP ≤ 100 mm Hg for falls was $P=0.024$.
††The P value of the interaction between treatment status and mean supine and standing SBP ≤ 100 mm Hg for syncope was $P=0.021$.

mortality. These associations were consistent across both treated and untreated groups, echoing prior research on supine and nocturnal hypertension as contributors to end-organ injury^{34–39} and affirming the potential harms of high BP even when standing. We also observed that postural drops in SBP (≥ 20 mm Hg) were associated with a higher risk of CHD and mortality. It is unclear whether these BP drops are a cause of cardiac injury, or secondary to neurological dysfunction (eg, insidiously developing Parkinsonism or cerebral dysregulation), or reflect existing subclinical disease (eg, endothelial dysfunction and thickened, less compliant vessels) that impairs one’s ability to regulate BP across body positions.^{7,40–42} However, these findings reinforce suggestions that more intensive BP treatment among adults with orthostatic hypotension may be important to reduce the risk of cardiovascular events and all-cause mortality, as shown in recent work.^{3,43} Sex-stratified analyses revealed stronger associations between orthostatic BP drops and adverse outcomes in females compared with males.

These differences may reflect sex-specific physiological responses to postural changes, such as greater reliance on cardiac output in females versus vasoconstriction in males.^{44,45} Hormonal influences, including vasodilatory effects of estrogen, may further contribute to these disparities, especially after menopause.^{46,47} Falls and syncope were associated with a broad spectrum of BP metrics. While traditionally associated with low BP and orthostatic hypotension, our findings suggest that both drops and increases in standing SBP and DBP were associated with higher fall and syncope risk, especially among untreated participants. These findings add to the literature, which has focused more on orthostatic hypotension than orthostatic hypertension with respect to falls and syncope.^{28,48–51} Mechanisms underlying orthostatic hypertension remain unclear but may involve exaggerated sympathetic activation and arterial stiffness.^{52–54} With respect to falls, it is possible that higher BP contributes directly to fall risk, as observed by others.^{55–57} Notably, this association

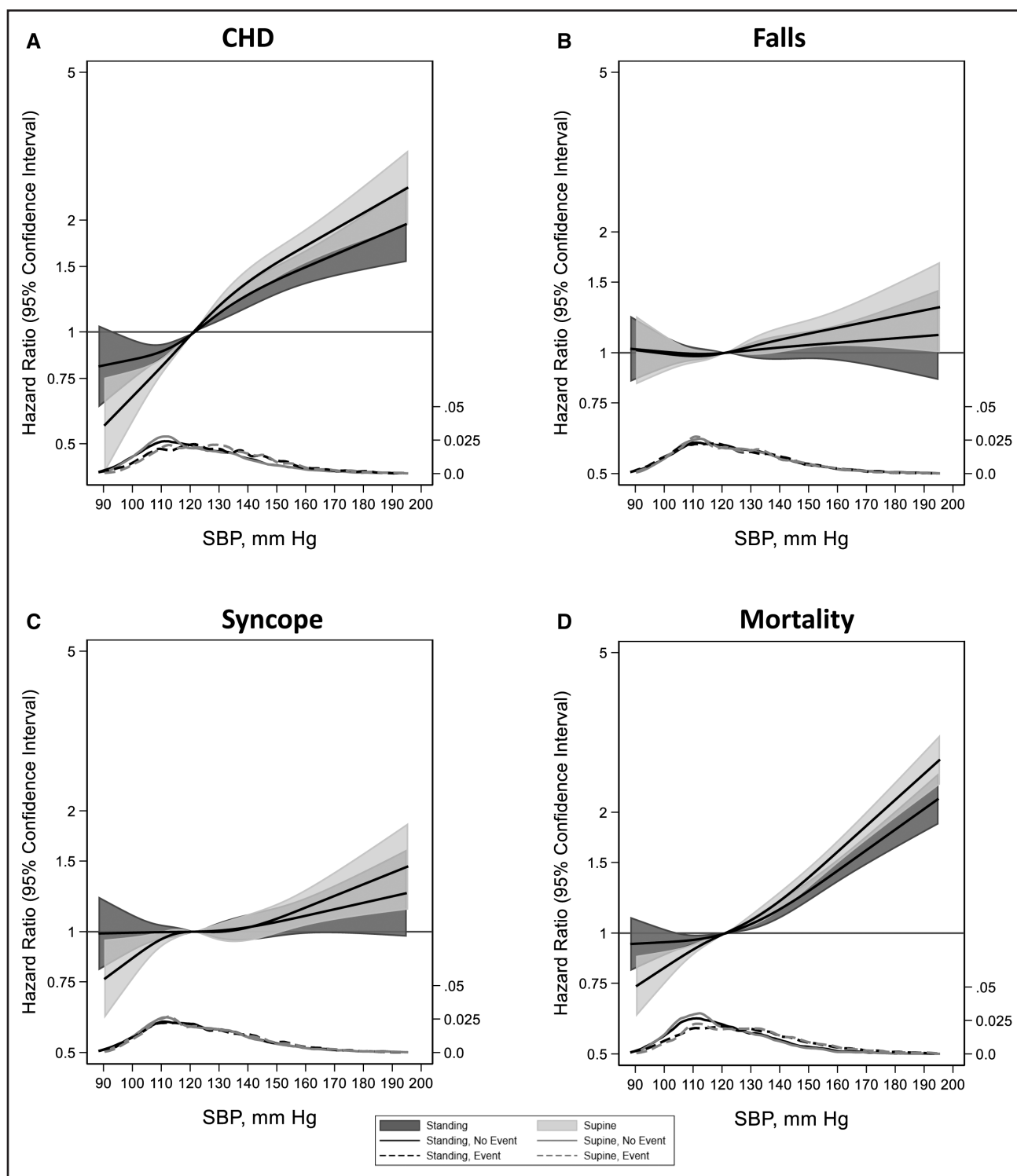


Figure. Adjusted association of supine and standing systolic blood pressure (SBP; mm Hg) modeled as restricted cubic splines.

Clinical outcomes were (A) coronary heart disease (CHD), (B) falls, (C) syncope, and (D) all-cause mortality. The association is modeled as restricted cubic splines (solid lines) with 4 knots determined by Harrell method. Dark gray shading indicates the 95% CI for standing SBP, while light gray shading indicates the 95% CI for supine SBP. All models used Cox proportional hazards models to determine hazard ratios, which are shown on a natural log scale. Adjustments were made for age, sex, race-center, estimated glomerular filtration rate, body mass index, heart rate, high-density lipoprotein cholesterol, total cholesterol, diabetes status, hypertension status, self-reported dizziness, alcohol consumption, education level, leisure index, smoking status, antihypertensive medication use in the past 2 weeks, and use of diuretics, antidepressants, sedatives, hypnotics, antipsychotics, and cholesterol-lowering medications. The figure display was truncated at the 0.5th and 99.5th percentiles of SBP. Kernel density plots depict the distribution of SBP by participants who experienced the outcome of interest (dashed line) vs those who did not (solid gray line).

Table 3. Composite Definitions and Risk of Cardiovascular and Hypotension-Related Outcomes Among Untreated Participants, N=8524 (HR, 95% CI)

	Proportion of the population, %	HR (95% CI), Coronary heart disease	HR (95% CI), Falls	HR (95% CI), Syncope	HR (95% CI), Mortality
Self-reported dizziness upon standing	9.1	1.11 (0.94–1.31)	1.20 (1.05–1.37)*	1.19 (1.02–1.37)*	1.08 (0.97–1.20)
Orthostatic change					
OH: Drop in SBP ≥ 20 mmHg or DBP ≥ 10 mmHg	2.8	1.94 (1.53–2.45)*	1.28 (1.00–1.63)	1.50 (1.17–1.92)*	1.74 (1.50–2.01)*†
OH: Drop in SBP ≥ 20 mmHg or DBP ≥ 5 mmHg	6.4	1.35 (1.14–1.61)*	1.17 (0.99–1.37)	1.41 (1.19–1.66)*	1.43 (1.28–1.60)*
OHTN: Increase in SBP ≥ 20 mmHg or DBP ≥ 10 mmHg	10.0	0.96 (0.80–1.14)	0.96 (0.83–1.10)	1.07 (0.93–1.24)‡	1.01 (0.91–1.13)
OHTN: Increase in SBP ≥ 20 mmHg or DBP ≥ 15 mmHg	2.5	0.87 (0.62–1.23)	1.37 (1.09–1.72)*	1.31 (1.02–1.68)*	1.05 (0.86–1.27)
High MAP ≥ 110 mmHg					
Supine	4.3	1.61 (1.30–1.98)*	1.17 (0.94–1.45)	1.34 (1.09–1.66)*	1.52 (1.34–1.73)*
Standing	6.1	1.39 (1.16–1.67)*	1.01 (0.83–1.22)	1.14 (0.94–1.37)	1.28 (1.14–1.44)*
Mean supine and standing	4.7	1.50 (1.22–1.84)*	1.09 (0.88–1.35)	1.26 (1.03–1.55)*	1.49 (1.31–1.69)*
Low MAP ≤ 70 mmHg					
Supine	2.9	0.68 (0.43–1.06)	1.01 (0.80–1.27)	0.75 (0.55–1.01)	0.81 (0.65–1.02)
Standing	1.9	0.59 (0.35–0.99)*	0.96 (0.72–1.28)	0.91 (0.64–1.28)	0.86 (0.66–1.12)
Mean supine and standing	2.0	0.62 (0.36–1.08)§	0.87 (0.65–1.15)	0.78 (0.55–1.12)	0.81 (0.62–1.06)
High pulse pressure ≥ 80 mmHg					
Supine	2.9	1.89 (1.48–2.41)*	0.92 (0.71–1.19)	1.23 (0.95–1.58)	1.71 (1.47–1.99)*
Standing	2.5	1.67 (1.26–2.20)*	0.96 (0.73–1.28)	1.26 (0.96–1.66)	1.60 (1.36–1.89)*
Mean supine and standing	2.5	1.87 (1.43–2.45)*	0.99 (0.74–1.32)	1.08 (0.81–1.45)	1.78 (1.52–2.10)*
Low pulse pressure ≤ 35 mmHg					
Supine	6.6	0.83 (0.67–1.03)	0.98 (0.82–1.16)	0.75 (0.61–0.92)*	0.85 (0.74–0.98)*
Standing	15.5	0.89 (0.77–1.03)	1.00 (0.89–1.12)	0.99 (0.87–1.12)¶	0.94 (0.86–1.04)
Mean supine and standing	9.1	0.79 (0.66–0.96)*	0.96 (0.82–1.11)	0.84 (0.71–0.99)*	0.87 (0.77–0.98)*
Hypertension: SBP ≥ 140 mmHg or DBP ≥ 90 mmHg					
Supine	15.2	1.66 (1.47–1.88)*	1.10 (0.98–1.24)	1.19 (1.05–1.35)*	1.42 (1.31–1.53)*
Standing	17.8	1.49 (1.32–1.68)*	1.06 (0.94–1.19)	1.14 (1.01–1.28)*	1.26 (1.17–1.37)*
Mean supine and standing	15.3	1.56 (1.37–1.76)*	1.09 (0.97–1.23)	1.15 (1.02–1.30)*	1.39 (1.28–1.51)*
Hypotension: SBP ≤ 105 mmHg or DBP ≤ 60 mmHg					
Supine	19.4	0.63 (0.54–0.74)*	0.85 (0.76–0.95)*#	0.85 (0.76–0.96)*	0.78 (0.71–0.86)*
Standing	17.9	0.78 (0.67–0.91)***	0.98 (0.88–1.09)	0.91 (0.81–1.03)††	0.92 (0.84–1.00)
Mean supine and standing	16.9	0.68 (0.58–0.81)*##	0.96 (0.86–1.08)	0.88 (0.77–1.00)§§	0.86 (0.78–0.95)*

Cox proportional hazards models with adjustment for age, race-study center, estimated glomerular filtration rate, body mass index, heart rate, high-density lipoprotein cholesterol, total cholesterol, diabetes, cholesterol-lowering medication use, alcohol use, education attainment, leisure activity index, and smoking status. DBP indicates diastolic blood pressure; HR, hazard ratio; HTN, orthostatic hypertension; MAP, mean arterial pressure; OH, orthostatic hypotension; and SBP, systolic blood pressure.

*Values were statistically significant with a $P < 0.05$.

†The P value of the interaction between treatment status and a drop in SBP ≥ 20 mmHg or DBP ≥ 5 mmHg for mortality was $P = 0.048$.

‡The P value of the interaction between treatment status and an increase in SBP ≥ 20 mmHg or DBP ≥ 10 mmHg for syncope was $P = 0.033$.

§The P value of the interaction between treatment status and low mean supine and standing MAP ≤ 70 mmHg for coronary heart disease was $P = 0.049$.

¶The P value of the interaction between treatment status and low standing pulse pressure ≤ 35 mmHg for syncope was $P = 0.039$.

#The P value of the interaction between treatment status and supine SBP ≤ 105 mmHg or DBP ≤ 60 mmHg for falls was $P = 0.004$.

**The P value of the interaction between treatment status and standing SBP ≤ 105 mmHg or DBP ≤ 60 mmHg for coronary heart disease was $P = 0.003$.

††The P value of the interaction between treatment status and standing SBP ≤ 105 mmHg or DBP ≤ 60 mmHg for syncope was $P = 0.002$.

##The P value of the interaction between treatment status and mean supine and standing SBP ≤ 105 mmHg or DBP ≤ 60 mmHg for coronary heart disease was $P = 0.011$.

§§The P value of the interaction between treatment status and mean supine and standing SBP ≤ 105 mmHg or DBP ≤ 60 mmHg for syncope was $P = 0.028$.

was attenuated among those receiving hypertension treatment. It is also possible that BP dysregulation itself affects balance.^{58–60} Unlike falls, high supine BP was also related to syncope. Studies have described greater

orthostatic changes among adults with higher resting BP.⁶¹ Moreover, chronic high BP exposure is thought to affect cerebral autoregulation, such that higher cerebral perfusion pressure is required to maintain cerebral

Table 4. Association of Different Orthostatic Systolic Blood Pressure Metrics With Cardiovascular and Hypotension-Related Outcomes Among Treated Adults, N=2862 (HR, 95% CI)

	Proportion of the population, %	HR (95% CI), Coronary heart disease	HR (95% CI), Falls	HR (95% CI), Syncope	HR (95% CI), Mortality
Self-reported dizziness upon standing	11.3	1.27 (1.03–1.58)*	1.20 (0.98–1.47)	1.27 (1.03–1.57)*	1.18 (1.02–1.36)*
Change (standing minus supine)					
Standing SBP drops ≥ 20 mm Hg†	7.0	1.82 (1.44–2.30)*	1.12 (0.84–1.48)	1.48 (1.14–1.92)*	1.46 (1.24–1.73)*‡
Standing SBP drops $\geq 15\%$	5.7	1.66 (1.28–2.17)*	1.02 (0.74–1.41)	1.43 (1.07–1.91)*	1.41 (1.17–1.70)*
Standing SBP increase ≥ 20 mm Hg†	2.7	0.63 (0.36–1.10)	0.98 (0.66–1.47)	0.86 (0.56–1.34)	1.16 (0.88–1.54)
Standing SBP increase $\geq 15\%$	3.2	0.70 (0.43–1.12)	1.12 (0.79–1.58)	0.93 (0.64–1.37)	0.99 (0.77–1.28)
High blood pressure					
Supine SBP ≥ 165 mm Hg	7.9	1.79 (1.41–2.27)*	1.17 (0.88–1.55)	1.21 (0.92–1.59)	1.96 (1.68–2.30)*
Standing SBP ≥ 165 mm Hg	7.9	1.53 (1.19–1.98)*	1.12 (0.85–1.48)	1.05 (0.80–1.38)	1.53 (1.30–1.80)*
Mean supine and standing SBP ≥ 165 mm Hg	7.4	1.57 (1.21–2.03)*	1.00 (0.74–1.34)	1.02 (0.76–1.36)	1.68 (1.43–1.99)*
Low blood pressure					
Supine SBP ≤ 100 mm Hg	2.7	0.91 (0.54–1.53)	1.37 (0.95–1.98)§	0.84 (0.52–1.36)	0.92 (0.65–1.31)
Standing SBP ≤ 100 mm Hg	4.7	1.18 (0.83–1.66)¶	1.21 (0.89–1.64)	1.55 (1.14–2.12)*#	1.13 (0.89–1.43)
Mean supine and standing SBP ≤ 100 mm Hg	2.7	0.69 (0.39–1.23)	1.42 (0.98–2.04)**	1.27 (0.84–1.93)††	0.91 (0.64–1.29)
Continuous (orthostatic change in standing)					
Change in SBP per 10 mm Hg decrease	...	0.86 (0.81–0.92)*	0.95 (0.90–1.01)	0.92 (0.87–0.98)*	0.89 (0.85–0.92)*
Percent change in SBP per 10 percentage points decrease	...	0.83 (0.76–0.90)*	0.94 (0.87–1.02)	0.89 (0.82–0.96)*	0.86 (0.82–0.91)*

Cox proportional hazards models with adjustment for age, race-study center, estimated glomerular filtration rate, body mass index, heart rate, high-density lipoprotein cholesterol, total cholesterol, diabetes, cholesterol-lowering medication use, alcohol use, education attainment, leisure activity index, and smoking status. HR indicates hazard ratio; and SBP, systolic blood pressure.

*Values were statistically significant with a $P < 0.05$.

†The 20 mm Hg value is used clinically to define postural change in blood pressure for both orthostatic hypotension and orthostatic hypertension. The other thresholds were chosen to be a similar proportion in the population as orthostatic hypotension ($\approx 7.0\%$) rounded to the nearest multiple of 5 for ease of clinical translation.

#The P value of the interaction between treatment status and standing SBP drops ≥ 20 mm Hg for mortality was $P = 0.035$.

§The P value of the interaction between treatment status and supine SBP ≤ 100 mm Hg for falls was $P = 0.013$.

¶The P value of the interaction between treatment status and standing SBP ≤ 100 mm Hg for coronary heart disease was $P = 0.004$.

#The P value of the interaction between treatment status and standing SBP ≤ 100 mm Hg for syncope was $P < 0.001$.

**The P value of the interaction between treatment status and mean supine and standing SBP ≤ 100 mm Hg for falls was $P = 0.024$.

††The P value of the interaction between treatment status and mean supine and standing SBP ≤ 100 mm Hg for syncope was $P = 0.021$.

blood flow.^{42,62} The combination of these effects (larger fluctuations in pressure and greater pressure requirements) may contribute to cerebral hypoperfusion after standing. In addition, there is evidence that supine hypertension can cause pressure natriuresis and worsen orthostatic tolerance in the morning.^{63,64} However, these mechanisms could not be evaluated in the present study.

Across outcomes and treatment strata, relative changes in SBP and DBP conferred similar risks as absolute BP changes. This finding has important implications. Orthostatic hypotension is less likely to be formally diagnosed among individuals with a lower supine or seated BP due to a smaller magnitude of absolute changes that may not meet the threshold definition of orthostatic hypotension.⁷ A relative drop in BP among those with lower or normal BP may be useful for identifying adults at higher risk for CHD, syncope, or death.⁷ Future work should explore whether those identified by relative change in BP, but not absolute changes, are

at greater risk for CVD and hypotension-related outcomes. Nonetheless, our data suggest that, for clinical or epidemiological applications, either approach can be used to characterize risk. This challenges prior hypotheses emphasizing relative change as a more physiologically meaningful metric.⁷ We also noted that the current thresholds of $\pm 20/10$ mm Hg for orthostatic hypotension or orthostatic hypertension are not ideally aligned. While SBP tends to stay constant or mildly drop upon standing, DBP tends to rise.⁶⁵ In the present study, alternate orthostatic definitions with the diastolic thresholds changed to a drop by ≥ 5 mm Hg for orthostatic hypotension and a rise by ≥ 15 mm Hg for orthostatic hypertension were more strongly associated with adverse events. This has been observed by others.⁶⁶ Given their alignment with underlying physiology, it may be useful to reconsider the 20/10 mm Hg consensus definitions if these findings are replicated in other studies.

Treatment status had little effect on the observed associations, with 1 notable exception. Low standing

Table 5. Composite Definitions and Risk of Cardiovascular and Hypotension-Related Outcomes Among Treated Participants, N=2862 (HR, 95% CI)

	Proportion of the population, %	HR (95% CI), Coronary heart disease	HR (95% CI), Falls	HR (95% CI), Syncope	HR (95% CI), Mortality
Self-reported dizziness upon standing	11.3	1.27 (1.03–1.58)*	1.20 (0.98–1.47)	1.27 (1.03–1.57)*	1.18 (1.02–1.36)*
Orthostatic change					
OH: Drop in SBP ≥ 20 mmHg or DBP ≥ 10 mmHg	7.6	1.84 (1.47–2.30)*	1.13 (0.86–1.48)	1.56 (1.22–2.00)*	1.44 (1.22–1.69)*†
OH: Drop in SBP ≥ 20 mmHg or DBP ≥ 5 mmHg	12.6	1.54 (1.27–1.85)*	1.09 (0.88–1.34)	1.51 (1.24–1.84)*	1.35 (1.18–1.54)*
OHTN: Increase in SBP ≥ 20 mmHg or DBP ≥ 10 mmHg	11.9	0.76 (0.59–0.98)*	0.90 (0.72–1.11)	0.81 (0.65–1.01)‡	0.92 (0.79–1.07)
OHTN: Increase in SBP ≥ 20 mmHg or DBP ≥ 15 mmHg	4.0	0.76 (0.50–1.15)	1.02 (0.73–1.42)	0.86 (0.60–1.24)	1.17 (0.93–1.48)
High MAP ≥ 110 mmHg					
Supine	10.2	1.49 (1.19–1.86)*	1.36 (1.07–1.74)*	1.23 (0.96–1.57)	1.67 (1.44–1.94)*
Standing	14.1	1.42 (1.16–1.74)*	1.06 (0.85–1.33)	1.09 (0.88–1.35)	1.46 (1.28–1.67)*
Mean supine and standing	11.1	1.40 (1.12–1.74)*	1.11 (0.87–1.43)	1.27 (1.01–1.60)*	1.55 (1.35–1.79)*
Low MAP ≤ 70 mmHg					
Supine	1.3	0.94 (0.44–2.00)	1.41 (0.85–2.35)	1.12 (0.61–2.05)	1.09 (0.67–1.76)
Standing	1.5	1.11 (0.64–1.93)	0.83 (0.47–1.48)	1.14 (0.64–2.02)	1.11 (0.77–1.61)
Mean supine and standing	0.9	1.30 (0.61–2.76)§	1.13 (0.60–2.13)	1.24 (0.63–2.40)	1.06 (0.62–1.80)
High pulse pressure ≥ 80 mmHg					
Supine	9.8	1.57 (1.25–1.97)*	1.02 (0.79–1.31)	1.14 (0.89–1.46)	1.79 (1.54–2.07)*
Standing	7.5	1.15 (0.88–1.50)	1.11 (0.85–1.45)	1.13 (0.86–1.48)	1.49 (1.27–1.76)*
Mean supine and standing	7.8	1.36 (1.05–1.76)*	1.01 (0.77–1.34)	1.00 (0.75–1.32)	1.64 (1.40–1.92)*
Low pulse pressure ≤ 35 mmHg					
Supine	2.9	0.69 (0.41–1.16)	1.02 (0.69–1.52)	0.94 (0.61–1.45)	0.85 (0.61–1.18)
Standing	9.7	1.03 (0.80–1.31)	1.19 (0.95–1.50)	1.27 (1.02–1.60)*¶	0.89 (0.75–1.06)
Mean supine and standing	4.8	0.89 (0.62–1.27)	1.20 (0.89–1.62)	1.13 (0.82–1.56)	0.81 (0.63–1.05)
Hypertension: SBP ≥ 140 mmHg or DBP ≥ 90 mmHg					
Supine	32.9	1.34 (1.15–1.56)*	1.08 (0.93–1.25)	1.15 (0.99–1.34)	1.37 (1.24–1.51)*
Standing	33.3	1.23 (1.06–1.44)*	1.13 (0.97–1.31)	1.10 (0.94–1.28)	1.23 (1.11–1.35)*
Mean supine and standing	32.1	1.26 (1.08–1.47)*	1.12 (0.96–1.30)	1.14 (0.98–1.33)	1.34 (1.21–1.48)*
Hypotension: SBP ≤ 105 mmHg or DBP ≤ 60 mmHg					
Supine	10.3	0.79 (0.60–1.04)	1.11 (0.90–1.37)‡	0.99 (0.78–1.25)	0.86 (0.73–1.03)
Standing	11.1	1.09 (0.86–1.37)**	1.14 (0.93–1.40)	1.31 (1.06–1.62)*††	1.02 (0.87–1.19)
Mean supine and standing	9.6	0.95 (0.73–1.23)‡‡	1.12 (0.90–1.39)	1.16 (0.91–1.47)§§	0.86 (0.72–1.03)

Cox proportional hazards models with adjustment for age, race-study center, estimated glomerular filtration rate, body mass index, heart rate, high-density lipoprotein cholesterol, total cholesterol, diabetes, cholesterol-lowering medication use, alcohol use, education attainment, leisure activity index, and smoking status. DBP indicates diastolic blood pressure; HR, hazard ratio; MAP, mean arterial pressure; OH, orthostatic hypotension; OHTN, orthostatic hypertension; and SBP, systolic blood pressure.

*Values were statistically significant with a $P < 0.05$.

†The P value of the interaction between treatment status and a drop in SBP ≥ 20 mmHg or DBP ≥ 5 mmHg for mortality was $P = 0.048$.

‡The P value of the interaction between treatment status and an increase in SBP ≥ 20 mmHg or DBP ≥ 10 mmHg for syncope was $P = 0.033$.

§The P value of the interaction between treatment status and low mean supine and standing MAP ≤ 70 mmHg for coronary heart disease was $P = 0.049$.

¶The P value of the interaction between treatment status and low standing pulse pressure ≤ 35 mmHg for syncope was $P = 0.039$.

#The P value of the interaction between treatment status and supine SBP ≤ 105 mmHg or DBP ≤ 60 mmHg for falls was $P = 0.004$.

**The P value of the interaction between treatment status and standing SBP ≤ 105 mmHg or DBP ≤ 60 mmHg for coronary heart disease was $P = 0.003$.

††The P value of the interaction between treatment status and standing SBP ≤ 105 mmHg or DBP ≤ 60 mmHg for syncope was $P = 0.002$.

‡‡The P value of the interaction between treatment status and mean supine and standing SBP ≤ 105 mmHg or DBP ≤ 60 mmHg for coronary heart disease was $P = 0.011$.

§§The P value of the interaction between treatment status and mean supine and standing SBP ≤ 105 mmHg or DBP ≤ 60 mmHg for syncope was $P = 0.028$.

SBP (≤ 100 mmHg) was associated with higher syncope risk only among treated individuals. Similarly, although not statistically significant, low SBP showed a higher risk of falls among the treated population compared with the untreated population. This suggests that low BP while

standing in the setting of hypertension treatment may be a higher risk state for hypotension-related adverse events. This is consistent with findings from the Systolic Intervention Trial (SPRINT), which found that more intensive BP treatment was associated with more frequent

syncope events, hypotensive episodes, and hypotension events.^{67,68} This is also in stark contrast to pulse pressure, which, while often cited as a reason to avoid treatment among older adults, was not associated with a higher risk for falls or syncope regardless of treatment status.^{69,70} This supports work in SPRINT and a recent meta-analysis, suggesting that more intensive hypertension treatment was similarly beneficial for hypertensive adults with a low DBP.^{68,71} Our findings suggest that BP assessments may provide complementary information to pulse pressure when evaluating syncope risk in the context of hypertension treatment. However, further research is needed to determine their clinical utility.

Our study has limitations. First, BP was assessed at a single baseline visit, which prevents us from evaluating the stability or evolution of orthostatic patterns over time. Second, because standing BP for participants who reported dizziness while standing was measured after participants felt safe to stand, we may have missed initial orthostatic hypotension and its potential rebound BP response which could lead to misclassification of orthostatic patterns. Third, we were unable to account for some pathophysiological modifiers of orthostatic BP response such as hydration status, physical conditioning, and sleep apnea which could affect both BP changes and risk for hypotension-related events, potentially introducing residual confounding. Similarly, our design cannot fully rule out the possibility of reverse causation, that is, that subclinical disease or comorbidities resulted in orthostatic hypotension or orthostatic hypertension. Fourth, although we stratified by antihypertensive treatment status, we lacked data on medication stability, adherence, and time since last dose which could affect acute BP changes especially standing BPs. While medication use was assessed at follow-up visits, orthostatic BP was only measured at baseline, limiting our ability to evaluate how changes in medication use over time may have influenced orthostatic BP responses or their associations with outcomes. Fifth, we analyzed BP metrics across standing and supine positions and did not include seated BP as a comparator to assess the relative performance of supine versus seated measures. This may be a useful focus of subsequent work. Sixth, supine BP was measured after a prolonged rest period (≈ 20 minutes), which may be challenging to implement in the clinic setting, but may be feasible at home. Seventh, heart rate was not recorded during standing, limiting our ability to assess baroreflex responses or autonomic compensation. Future studies incorporating concurrent heart rate and BP measurements during postural changes could help clarify underlying mechanisms. Eighth, falls and syncope events were ascertained from *International Classification of Diseases, Ninth Revision* and *International Classification of Diseases, Tenth Revision* codes within administrative claims data, which may underdetect less severe events that did not result in hospitalization or require medical attention. Ninth, we did not have

concurrent overnight urine data to assess for the role of pressure natriuresis as a mechanism of syncope or falls. Tenth, given the number of comparisons in this study, the possibility of false-positive findings cannot be excluded. Finally, the generalizability of our results may be limited to middle-aged, ambulatory, community-dwelling Black and White adults and may not extend to older adults with greater frailty or institutionalized populations.

Despite these limitations, our study also has strengths. We used data from a well-characterized, longitudinal cohort with standardized BP measurement protocols and rigorously adjudicated cardiovascular and mortality outcomes. The large sample size allowed for precise estimation of associations across BP phenotypes, including orthostatic hypertension, which remains underexamined in epidemiological literature. Our assessment of hemodynamic profiles and their prognostic relevance was comprehensive, including both traditional (SBP and DBP) and derived BP metrics (MAP and pulse pressure), as well as absolute and relative orthostatic changes. Finally, our stratification by concurrent antihypertensive treatment allowed us to explore treatment-modifying effects on the associations between positional BPs and adverse cardiovascular events.

Our findings have clinical implications. Both larger drops and increases in BP upon standing, rather than low absolute BP values alone, may serve as potential markers of autonomic dysfunction and vascular pathology. Moreover, BP measurement in the seated position alone may miss important BP changes associated with CVD or hypotension-related events. While we did not observe consistent harmful associations between antihypertensive treatment and adverse events in the context of orthostatic changes, these findings highlight the potential value of characterizing BP patterns before modifying therapy. Nonetheless, future studies should define reproducible thresholds for orthostatic hypotension and hypertension and evaluate whether targeted interventions based on positional BP profiles may improve clinical outcomes in diverse populations.

In conclusion, in this large cohort of middle-aged Black and White adults, both absolute and relative BP drops and increases were associated with adverse cardiovascular and hypotension-related outcomes. Orthostatic hypertension was associated with falls and syncope among untreated participants, and standing hypotension was associated with syncope among treated participants. These findings may inform future research seeking to optimize hypertension treatment strategies that balance CVD risk reduction with the prevention of hypotension-related complications.

PERSPECTIVES

This study advances knowledge regarding orthostatic BP regulation and its association with adverse cardiovascular and hypotension-related events among middle-aged

adults. Our data show that both drops and increases in BP upon standing are consistently associated with cardiovascular and hypotension-related events, suggesting a shared pathophysiology in which underlying autonomic and vascular dysfunction, not evident from seated and supine measurements alone, contributes to end-organ injury and postural instability.

These findings highlight the potential value of positional BP assessments to better characterize risk as seated BP measurements alone may miss clinically relevant patterns. The observed associations between supine and orthostatic hypertension with CHD, syncope, and all-cause mortality raise important questions about whether these BP variations represent markers of subclinical disease or modifiable risk states.

Future research should focus on investigating novel pathophysiological pathways linking orthostatic BP changes to adverse outcomes and evaluate whether treatment strategies informed by orthostatic BP patterns improve cardiovascular and safety outcomes.

ARTICLE INFORMATION

Received August 2, 2025; accepted May 5, 2026.

An abstract of this work was presented at the AHA EPI|Lifestyle Conference in New Orleans, LA, in March 2025.

Affiliations

Division of General Medicine, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA (M.N.M.-K., M.Z., R.-A.N.T.-O., F.L.K., S.P.J.). University of Mississippi Medical Center, Memory Impairment and Neurodegenerative Dementia Center, Department of Medicine, Jackson, MS (B.G.W.). Department of Epidemiology, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD (E.S.). Division of Epidemiology and Community Health, University of Minnesota, Minneapolis, MN (P.L.L.).

Acknowledgments

The ARIC study (Atherosclerosis Risk in Communities) has been funded in whole or in part with Federal funds from the National Heart, Lung, and Blood Institute, National Institutes of Health, Department of Health and Human Services, under Contract nos. (75N92022D00001, 75N92022D00002, 75N92022D00003, 75N92022D00004, 75N92022D00005). The authors thank the staff and participants of the ARIC study for their important contributions. For a full list of contributors to ARIC, please visit sites.csc.unc.edu/aric/. The principal investigator had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Sources of Funding

S.P. Juraschek was supported by National Institutes of Health (NIH)/National Heart, Lung, and Blood Institute (NHLBI) grant R01HL153191. E. Selvin was supported by NIH/NHLBI grant K24 HL152440. P.L. Lutsey was supported by NIH/NHLBI grant K24 HL159246. Dr Windham was supported by NIH/National Institute on Aging (NIA) grant R01 AG054787. The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the article; and decision to submit the article for publication.

Disclosures

None.

REFERENCES

- Fuchs FD, Whelton PK. High blood pressure and cardiovascular disease. *Hypertension*. 2020;75:285–292. doi: 10.1161/HYPERTENSIONAHA.119.14240
- Herrett E, Strongman H, Gadd S, Tomlinson L, Nitsch D, Bhaskaran K, Williamson E, van Staa T, Sofat R, Timmis A, et al. The importance of blood pressure thresholds versus predicted cardiovascular risk on subsequent rates of cardiovascular disease: a cohort study in English primary care. *Lancet Healthy Longev*. 2022;3:e22–e30. doi: 10.1016/S2666-7568(21)00281-6
- Juraschek SP, Taylor AA, Wright JT, Evans GW, Miller ER, Plante TB, Cushman WC, Gure TR, Haley WE, Moinuddin I, et al; SPRINT Research Group. Orthostatic hypotension, cardiovascular outcomes, and adverse events: results from SPRINT. *Hypertension*. 2020;75:660–667. doi: 10.1161/HYPERTENSIONAHA.119.14309
- Mezue K, Goyal A, Pressman GS, Matthew R, Horrow JC, Rangaswami J. Blood pressure variability predicts adverse events and cardiovascular outcomes in SPRINT. *J Clin Hypertens (Greenwich)*. 2018;20:1247–1252. doi: 10.1111/jch.13346
- Jiang X, Mang X, Zhou H, Chen J, Tan H, Ren H, Huang B, Zhong L, Lipsitz LA, Manor B, et al. The physiologic complexity of beat-to-beat blood pressure is associated with age-related alterations in blood pressure regulation. *Aging Cell*. 2024;23:e13943. doi: 10.1111/ace1.13943
- Lagro J, Schoon Y, Heerts I, Meel-van den Abeelen ASS, Schalk B, Wieling W, Olde Rikkert MGM, Claassen JAHR. Impaired systolic blood pressure recovery directly after standing predicts mortality in older falls clinic patients. *J Gerontol A Biol Sci Med Sci*. 2014;69:471–478. doi: 10.1093/gerona/glt111
- Juraschek SP, Cortez MM, Flack JM, Ghazi L, Kenny RA, Rahman M, Spikes T, Shibao CA, Biaggioni I; American Heart Association Council on Hypertension. Orthostatic hypotension in adults with hypertension: a scientific statement from the American Heart Association. *Hypertension*. 2024;81:e16–e30. doi: 10.1161/HYP.0000000000000236
- Melgarejo JD, Yang WY, Thijs L, Li Y, Asayama K, Hansen TW, Wei F-F, Kikuya M, Ohkubo T, Dolan E, et al; International Database on Ambulatory Blood Pressure in Relation to Cardiovascular Outcome Investigators. Association of fatal and nonfatal cardiovascular outcomes with 24-hour mean arterial pressure. *Hypertension*. 2021;77:39–48. doi: 10.1161/HYPERTENSIONAHA.120.14929
- Hussain SM, Ernst ME, Barker AL, Margolis KL, Reid CM, Neumann JT, Tonkin AM, Phuong Thao Le Thi, Beilin LJ, Pham T, et al. Variation in mean arterial pressure increases falls risk in elderly physically frail and pre-frail individuals treated with antihypertensive medication. *Hypertension*. 2022;79:2051–2061. doi: 10.1161/HYPERTENSIONAHA.122.19356
- Maruhashi T, Kajikawa M, Kishimoto S, Yamaji T, Harada T, Hashimoto Y, Mizobuchi A, Tanigawa S, Yusoff FM, Nakano Y, et al. Percentage of mean arterial pressure as a marker of atherosclerosis for detecting patients with coronary artery disease. *Hypertens Res*. 2024;47:281–290. doi: 10.1038/s41440-023-01442-4
- McEvoy JW, Chen Y, Rawlings A, Hoogeveen RC, Ballantyne CM, Blumenthal RS, Coresh J, Selvin E. Diastolic blood pressure, subclinical myocardial damage, and cardiac events: implications for blood pressure control. *J Am Coll Cardiol*. 2016;68:1713–1722. doi: 10.1016/j.jacc.2016.07.754
- Van Bortel LMAB, Struijker-Boudier HAJ, Safar ME. Pulse pressure, arterial stiffness, and drug treatment of hypertension. *Hypertension*. 2001;38:914–921. doi: 10.1161/hy1001.095773
- Farnett L, Mulrow CD, Linn WD, Lucey CR, Tuley MR. The J-curve phenomenon and the treatment of hypertension. Is there a point beyond which pressure reduction is dangerous? *JAMA*. 1991;265:489–495. doi: 10.1001/jama.1991.03460040065031
- Sarwal A, Boucher RE, Hartsell SE, Wei G, Shen J, Chertow GM, Whelton PK, Cheung AK, McEvoy JW, Greene T, et al. Baseline diastolic BP and BP-lowering effects on cardiovascular outcomes and all-cause mortality: a meta-analysis. *J Am Soc Nephrol*. 2025;36:911–922. doi: 10.1681/ASN.0000000539
- Beddhu S, Chertow GM, Cheung AK, Cushman WC, Rahman M, Greene T, Wei G, Campbell RC, Conroy M, Freedman BI, et al; SPRINT Research Group. Influence of baseline diastolic blood pressure on effects of intensive compared with standard blood pressure control. *Circulation*. 2018;137:134–143. doi: 10.1161/CIRCULATIONAHA.117.030848
- Manta E, Thomopoulos C, Kariori M, Polyzos D, Mihos C, Konstantinidis D, Farmakis D, Mancia G, Tsoufis K. Revisiting cardiovascular benefits of blood pressure reduction in primary and secondary prevention: focus on targets and residual risk—a systematic review and meta-analysis. *Hypertension*. 2024;81:1076–1086. doi: 10.1161/HYPERTENSIONAHA.123.22610
- Ambrož M, de Vries ST, Hoogenberg K, Denig P. Older age, polypharmacy, and low systolic blood pressure are associated with more hypotension-related adverse events in patients with type 2 diabetes treated with antihypertensives. *Front Pharmacol*. 2021;12:728911. doi: 10.3389/fphar.2021.728911

18. Egan BM, Kai B, Wagner CS, Fleming DO, Henderson JH, Chandler AH, Sinopoli A. Low blood pressure is associated with greater risk for cardiovascular events in treated adults with and without apparent treatment-resistant hypertension. *J Clin Hypertens (Greenwich)*. 2017;19:241–249. doi: 10.1111/jch.12904
19. BioLINCC: Atherosclerosis Risk in Communities Study (ARIC). Accessed July 18, 2025. <https://biolincc.nhlbi.nih.gov/studies/aric/>
20. White AD, Folsom AR, Chambless LE, Sharret AR, Yang K, Conwill D, Higgins M, Williams OD, Tyroler HA. Community surveillance of coronary heart disease in the Atherosclerosis Risk in Communities (ARIC) Study: methods and initial two years' experience. *J Clin Epidemiol*. 1996;49:223–233. doi: 10.1016/0895-4356(95)00041-0
21. Wright JD, Folsom AR, Coresh J, Sharrett AR, Couper D, Wagenknecht LE, Mosley TH, Ballantyne CM, Boerwinkle EA, Rosamond WD, et al. The ARIC (Atherosclerosis Risk in Communities) study: JACC focus seminar 3/8. *J Am Coll Cardiol*. 2021;77:2939–2959. doi: 10.1016/j.jacc.2021.04.035
22. Nardo CJ, Chambless LE, Light KC, Rosamond WD, Sharrett AR, Tell GS, Heiss G. Descriptive epidemiology of blood pressure response to change in body position. The ARIC study. *Hypertension*. 1999;33:1123–1129. doi: 10.1161/01.hyp.33.5.1123
23. Freeman R, Wieling W, Axelrod FB, Benditt DG, Benarroch E, Biaggioni I, Cheshire WP, Chelmsky T, Cortelli P, Gibbons CH, et al. Consensus statement on the definition of orthostatic hypotension, neurally mediated syncope and the postural tachycardia syndrome. *Clin Auton Res*. 2011;21:69–72. doi: 10.1007/s10286-011-0119-5
24. Niemelä V, Siddiqui F, Ameloot K, Reinikainen M, Grand J, Hästbacka J, Hassager C, Kjaergaard J, Åneman A, Tiainen M, et al. Higher versus lower blood pressure targets after cardiac arrest: systematic review with individual patient data meta-analysis. *Resuscitation*. 2023;189:109862. doi: 10.1016/j.resuscitation.2023.109862
25. Selvaraj S, Steg PG, Elbez Y, Sorbets E, Feldman LJ, Eagle KA, Magnus Ohman E, Blacher J, Bhatt DL; REACH Registry Investigators, et al. Pulse pressure and risk for cardiovascular events in patients with atherothrombosis. *JACC*. 2016;67:392–403. doi: 10.1016/j.jacc.2015.10.084
26. Franklin SS, Wong ND. Hypertension and cardiovascular disease: contributions of the framingham heart study. *Glob Heart*. 2013;8:49–57. doi: 10.1016/j.gheart.2012.12.004
27. Earle WB, Kondo JK, Kendrick KN, Turkson-Ocran R-A, Ngo L, Cluett J, Mukamal KJ, Daya Malek N, Selvin E, Lutsey PL, et al. Association of supine hypertension versus standing hypotension with adverse events among middle-aged Adults. *Hypertension*. 2023;80:2437. doi: 10.1161/HYPERTENSIONAHA.123.21215
28. Juraschek SP, Daya N, Appel LJ, Miller ER, Windham BG, Pompeii L, Griswold ME, Kucharska-Newton A, Selvin E. Orthostatic Hypotension in middle-age and risk of falls. *Am J Hypertens*. 2017;30:188–195. doi: 10.1093/ajh/hpw108
29. Whelton PK, Carey RM, Aronow WS, Casey DE, Collins KJ, Dennison Himmelfarb C, DePalma SM, Gidding S, Jamerson KA, Jones DW, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension*. 2018;71:e13–e115. doi: 10.1161/HYP.0000000000000065
30. Khan SS, Matsushita K, Sang Y, Ballew SH, Grams ME, Surapaneni A, Blaha MJ, Carson AP, Chang AR, Ciemins E, et al; Chronic Kidney Disease Prognosis Consortium and the American Heart Association Cardiovascular-Kidney-Metabolic Science Advisory Group. Development and validation of the American Heart Association's PREVENT Equations. *Circulation*. 2024;149:430–449. doi: 10.1161/CIRCULATIONAHA.123.067626
31. Inker LA, Eneanya ND, Coresh J, Tighiouart H, Wang D, Sang Y, Crews DC, Doria A, Estrella MM, Froissart M, et al; Chronic Kidney Disease Epidemiology Collaboration. New creatinine- and cystatin c-based equations to estimate GFR without race. *N Engl J Med*. 2021;385:1737–1749. doi: 10.1056/NEJMoa2102953
32. Richardson MT, Ainsworth BE, Wu HC, Jacobs DR, Leon AS. Ability of the Atherosclerosis Risk in Communities (ARIC)/Baecke Questionnaire to assess leisure-time physical activity. *Int J Epidemiol*. 1995;24:685–693. doi: 10.1093/ije/24.4.685
33. Harrell FE, Lee KL, Pollock BG. Regression models in clinical studies: determining relationships between predictors and response. *J Natl Cancer Inst*. 1988;80:1198–1202. doi: 10.1093/jnci/80.15.1198
34. Earle WB, Kondo JK, Kendrick KN, Turkson-Ocran R-A, Ngo L, Cluett JL, Mukamal KJ, Daya Malek N, Selvin E, Lutsey PL, et al. Association of supine hypertension versus standing hypotension with adverse events among middle-aged adults. *Hypertension*. 2023;80:2437–2446. doi: 10.1161/HYPERTENSIONAHA.123.21215
35. Wang F, Zhao H, Yang C, Kong G, Song L, Li C, Wang Y, Chen S, Wang J, Wu S. Association of blood pressure in the supine position with target organ damage in subjects over 60 years old. *J Int Med Res*. 2017;45:123–133. doi: 10.1177/0300060516677175
36. Palma JA, Redel-Traub G, Porciuncula A, Samaniego-Toro D, Millar Vernetti P, Lui YW, Norcliffe-Kaufmann L, Kaufmann H. The impact of supine hypertension on target organ damage and survival in patients with synucleinopathies and neurogenic orthostatic hypotension. *Parkinsonism Relat Disord*. 2020;75:97–104. doi: 10.1016/j.parkreldis.2020.04.011
37. Arnold AC, Biaggioni I. Management approaches to hypertension in autonomic failure. *Curr Opin Nephrol Hypertens*. 2012;21:481–485. doi: 10.1097/MNH.0b013e328356c52f
38. Jordan J, Biaggioni I. Diagnosis and treatment of supine hypertension in autonomic failure patients with orthostatic hypotension. *J Clin Hypertens (Greenwich)*. 2002;4:139–145. doi: 10.1111/j.1524-6175.2001.00516.x
39. Gao DM, Col H, Larbi Kwabong F, Turkson-Ocran RA, Ngo LH, Cluett JL, Wagenknecht L, Windham BG, Selvin E, Lutsey PL, et al. Supine blood pressure and risk of cardiovascular disease and mortality. *JAMA Cardiol*. 2025;10:265–275. doi: 10.1001/jamacardio.2024.5213
40. Zambach C, Fedorowski A, Gerward S, Johansson M, Engström G, Hamrefors V. Subclinical atherosclerosis and risk factors in relation to autonomic indices in the general population. *J Hypertens*. 2023;41:759–767. doi: 10.1097/HJH.0000000000003397
41. Palma JA, Kaufmann H. Orthostatic hypotension in Parkinson disease. *Clin Geriatr Med*. 2020;36:53–67. doi: 10.1016/j.cger.2019.09.002
42. Ruland S, Aiyagari V. Cerebral autoregulation and blood pressure lowering. *Hypertension*. 2007;49:977–978. doi: 10.1161/HYPERTENSIONAHA.107.087502
43. Juraschek SP, Hu JR, Cluett JL, Ishak AM, Mita C, Lipsitz LA, Appel LJ, Beckett NS, Coleman RL, Cushman WC, et al. Orthostatic hypotension. Hypertension treatment, and cardiovascular disease: an individual participant meta-analysis. *JAMA*. 2023;330:1459–1471. doi: 10.1001/jama.2023.18497
44. Patel K, Rössler A, Lackner HK, Trozic I, Laing C, Lorr D, Green DA, Hinghofer-Szalkay H, Goswami N. Effect of postural changes on cardiovascular parameters across gender. *Medicine (Baltimore)*. 2016;95:e4149. doi: 10.1097/MD.000000000000149
45. Fitzgibbon-Collins LK, Pereira TJ, Edgell H. Women, orthostatic tolerance, and POTS: a narrative review. *Auton Neurosci*. 2025;259:103284. doi: 10.1016/j.autneu.2025.103284
46. Guetta V, Cannon RO. Cardiovascular effects of estrogen and lipid-lowering therapies in postmenopausal women. *Circulation*. 1996;93:1928–1937. doi: 10.1161/01.cir.93.10.1928
47. Raj A, Chakole S, Agrawal S, Gupta A, Khekade H, Prasad R, Lohakare T, Wanjar M. The impact of menopause on cardiovascular aging: a comprehensive review of androgen influences. *Cureus*. 2023;15:e43569. doi: 10.7759/cureus.43569
48. Wahba A, Shibao CA, Muldowney JAS, Peltier A, Habermann R, Biaggioni I. Management of orthostatic hypotension in the hospitalized patient: a narrative review. *Am J Med*. 2022;135:24–31. doi: 10.1016/j.amjmed.2021.07.030
49. Mol A, Bui Hoang PTS, Sharmin S, Reijniers EM, van Wezel RJA, Meskers CGM, Maier AB. Orthostatic hypotension and falls in older adults: a systematic review and meta-analysis. *J Am Med Dir Assoc*. 2019;20:589–597.e5. doi: 10.1016/j.jamda.2018.11.003
50. Atkins D, Hanusa B, Sefcik T, Kapoor W. Syncope and orthostatic hypotension. *Am J Med*. 1991;91:179–185. doi: 10.1016/0002-9343(91)90012-m
51. Mussi C, Ungar A, Salvioli G, Menozzi C, Bartoletti A, Giada F, Lagi A, Ponassi I, Re G, Furlan R, et al; Evaluation of Guidelines in Syncope Study 2 Group. Orthostatic hypotension as cause of syncope in patients older than 65 years admitted to emergency departments for transient loss of consciousness. *J Gerontol A Biol Sci Med Sci*. 2009;64:801–806. doi: 10.1093/gerona/glp028
52. Tromp J, Shen L, Jhund PS, Anand IS, Carson PE, Desai AS, Granger CB, Komajda M, McKelvie RS, Pfeffer MA, et al. Age-related characteristics and outcomes of patients with heart failure with preserved ejection fraction. *J Am Coll Cardiol*. 2019;74:601–612. doi: 10.1016/j.jacc.2019.05.052
53. Fessel J, Robertson D. Orthostatic hypertension: when pressor reflexes overcompensate. *Nat Clin Pract Nephrol*. 2006;2:424–431. doi: 10.1038/ncpneph0228
54. Kario K. Morning surge in blood pressure and cardiovascular risk: evidence and perspectives. *Hypertension*. 2010;56:765–773. doi: 10.1161/HYPERTENSIONAHA.110.157149
55. Welmer AK, Wang R, Rizzuto D, Ek S, Vetrano DL, Qiu C. Associations of blood pressure with risk of injurious falls in old age vary by

- functional status: a cohort study. *Exp Gerontol.* 2020;140:111038. doi: 10.1016/j.exger.2020.111038
56. Abu Bakar AAZ, Abdul Kadir A, Idris NS, Mohd Nawi SN. Older adults with hypertension: prevalence of falls and their associated factors. *Int J Environ Res Public Health.* 2021;18:8257. doi: 10.3390/ijerph18168257
 57. Swiatoniowska-Lonc N, Jaciow E, Polanski J, Jankowska-Polanska B. Falls among patients with hypertension - preliminary study. *Eur J Prev Cardiol.* 2021;28(Supplement_1):zwab061.413. doi: 10.1093/eurjpc/zwab061.413.
 58. Lopes AR, Moreira MD, Trelha CS, Marchiori LL de M. Association between complaints of dizziness and hypertension in non-institutionalized elders. *Int Arch Otorhinolaryngol.* 2013;17:157–162. doi: 10.7162/S1809-97772013000200007
 59. Gangavati A, Hajjar I, Quach L, Jones RN, Kiely DK, Gagnon P, Lipsitz LA. Hypertension, orthostatic hypotension, and the risk of falls in a community-dwelling elderly population: the maintenance of balance, independent living, intellect, and zest in the elderly of Boston study. *J Am Geriatr Soc.* 2011;59:383–389. doi: 10.1111/j.1532-5415.2011.03317.x
 60. Acar S, Demirbüken I, Algun C, Malkoç M, Tekin N. Is hypertension a risk factor for poor balance control in elderly adults? *J Phys Ther Sci.* 2015;27:901–904. doi: 10.1589/jpts.27.901
 61. Roca F, Rougette K, Zmuda L, Noel G, Larose S, Bordage M, Chassagne P. Association between orthostatic blood pressure dysregulation and geriatric syndromes: a cross-sectional study. *BMC Geriatr.* 2022;22:157. doi: 10.1186/s12877-022-02844-8
 62. Strandgaard S, Paulson OB. Cerebral autoregulation. *Stroke.* 1984;15:413–416. doi: 10.1161/01.str.15.3.413
 63. Uzu T, Takeji M, Yamauchi A, Kimura G. Circadian rhythm and postural change in natriuresis in non-dipper type of essential hypertension. *J Hum Hypertens.* 2001;15:323–327. doi: 10.1038/sj.jhh.1001185
 64. Okamoto LE, Gamboa A, Shibao C, Black BK, Diedrich A, Raj SR, Robertson D, Biaggioni I. Nocturnal blood pressure dipping in the hypertension of autonomic failure. *Hypertension.* 2009;53:363–369. doi: 10.1161/HYPERTENSIONAHA.108.124552
 65. Jordan J, Ricci F, Hoffmann F, Hamrefors V, Fedorowski A. Orthostatic hypertension: critical appraisal of an overlooked condition. *Hypertension.* 2020;75:1151–1158. doi: 10.1161/HYPERTENSIONAHA.120.14340
 66. Palatini P. 'Orthostatic hypertension: a harbinger of adverse cardiovascular outcomes in young individuals'. *Eur J Prev Cardiol.* 2024;31:e105. doi: 10.1093/eurjpc/zwad249
 67. Zhang W, Redline S, Viswanathan A, Ascher SB, Hari D, Juraschek SP, Tzourio C, Drawz PE, Lipsitz LA, Mittleman MA, et al. Hypotensive episodes on 24-hour ambulatory blood pressure and cognitive function: insights from the SPRINT study. *Hypertension.* 2025;82:627–637. doi: 10.1161/HYPERTENSIONAHA.124.24222
 68. Research Group SPRINT, Wright JT, Williamson JD, Whelton PK, Snyder JK, Sink KM, Rocco MV, Reboussin DM, Rahman M, Oparil S, Lewis CE, et al. A randomized trial of intensive versus standard blood-pressure control. *N Engl J Med.* 2015;373:2103–2116. doi: 10.1056/NEJMoa1511939
 69. Pinto E. Blood pressure and ageing. *Postgrad Med J.* 2007;83:109–114. doi: 10.1136/pgmj.2006.048371
 70. Tannenbaum J, Bittner V, Waters DD. When diastole lets you down: clinical relevance of a widened pulse pressure. *Can J Cardiol.* 2020;36:593–595. doi: 10.1016/j.cjca.2019.10.028
 71. Juraschek SP, Hu JR, Cluett JL, Mita C, Lipsitz LA, Appel LJ, Beckett NS, Davis BR, Holman RR, Miller ER, et al. Effects of intensive blood pressure treatment on orthostatic hypertension: individual level meta-analysis. *BMJ.* 2025;388:e080507. doi: 10.1136/bmj-2024-080507